

Introduction

Modern cloud-native applications are increasingly deployed on Kubernetes, a powerful container orchestration platform trusted by organisations worldwide to run critical, internet-facing services. However, as adoption grows, so does the attack surface. Misconfigured pods, vulnerable containers, and undetected runtime anomalies represent real security risks that are difficult to identify without constant vigilance.

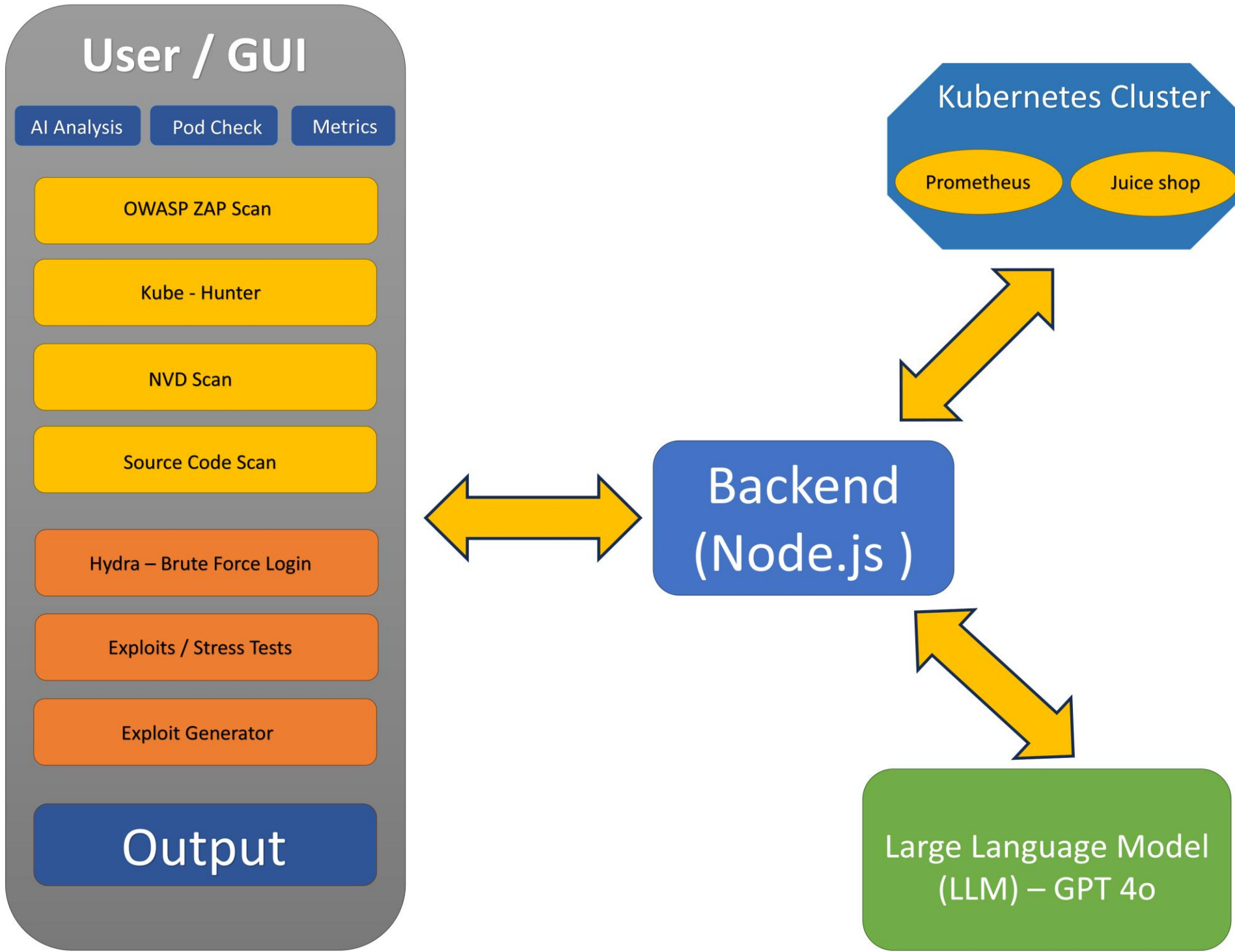
The Kubernetes AI Security Monitor addresses theses risks by applying Large Language Model (LLM) intelligence to real-time cluster data. The tool automatically collects pod health status, performance metrics, and live logs from a running Kubernetes cluster, then uses AI to analyse this data and produce a clear, structured security and health report.

As Kubernetes becomes the backbone of modern application infrastructure, the ability to rapidly detect and understand security-relevant events is critical. This project demonstrates how AI can augment security monitoring, reducing response times, lowering the expertise barrier, and making cluster security visibility accessible to teams and projects of all sizes.

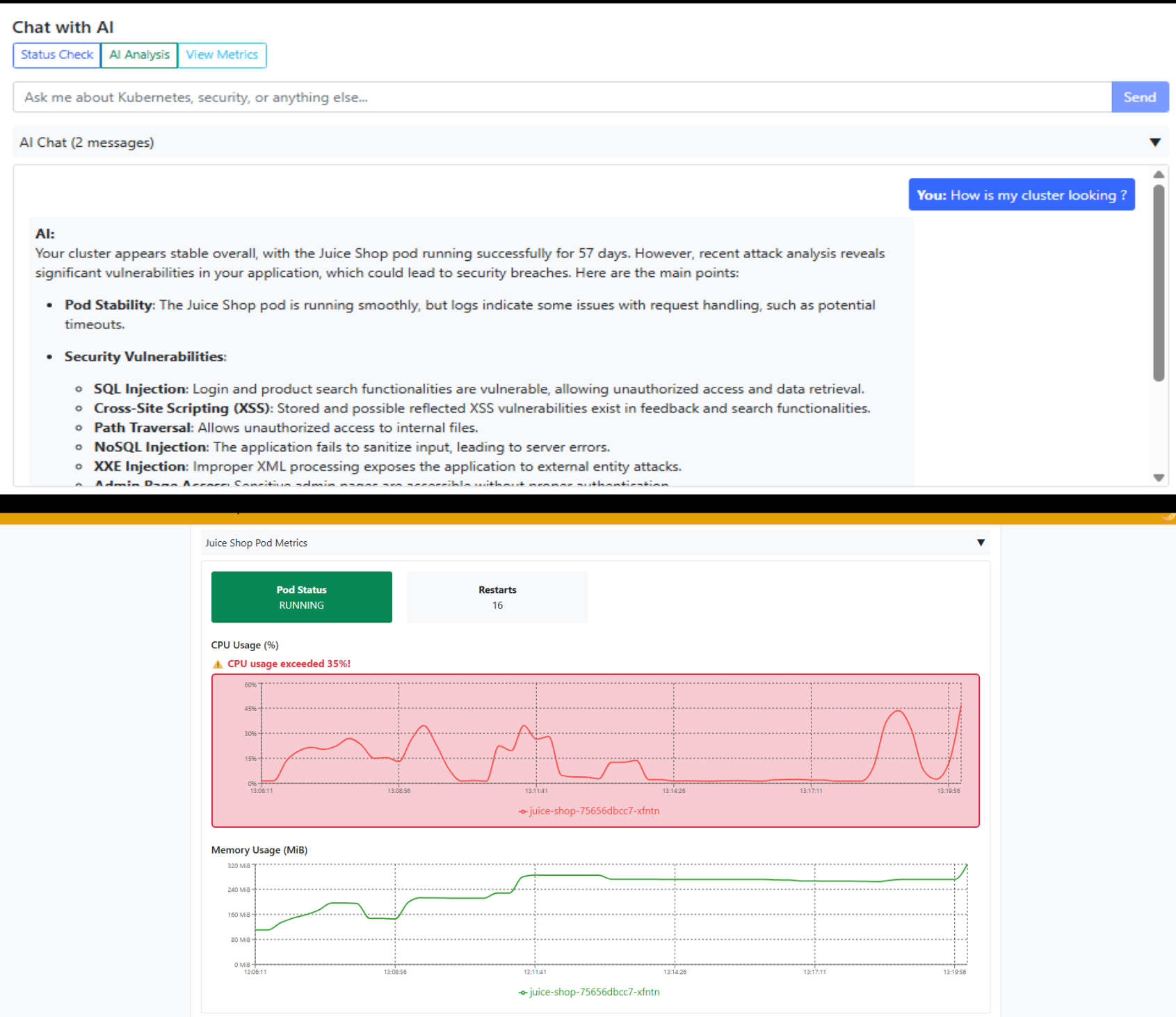
Core Technologies

- Kubernetes
 - Docker
 - JavaScript
 - React
- Helm
 - Cloud Server (AWS)
 - LLM Analysis
 - OWASP – Juice Shop

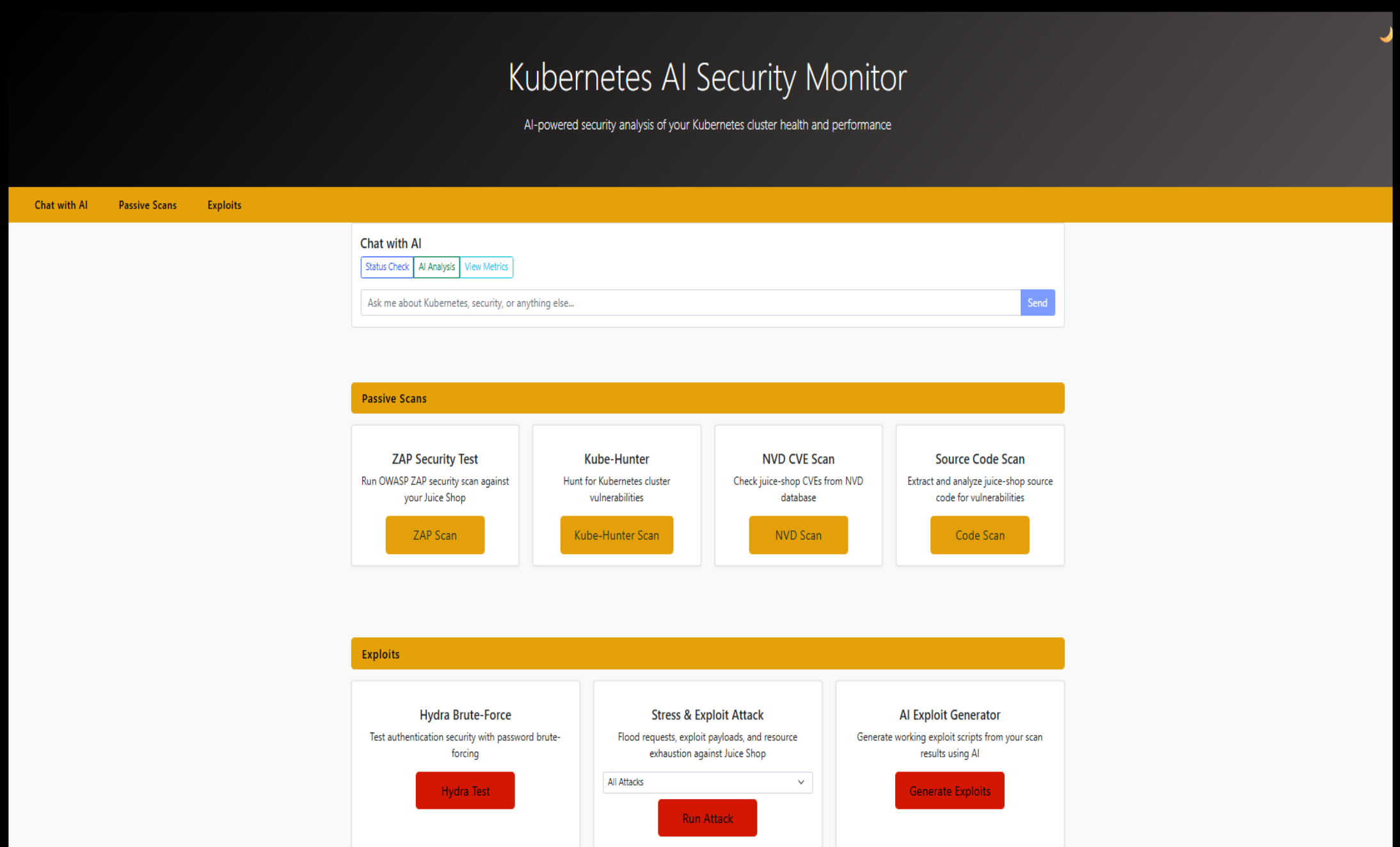
Architect Diagram



Real Time Analysis



User Interface



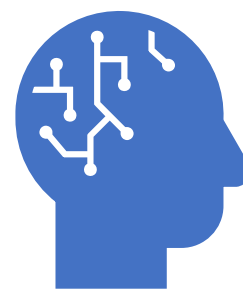
Security Features



Defence: Monitors live pod health and logs, using an LLM to generate timestamped security reports with findings and remediation recommendations.



Real-Time Metrics: Streams live CPU, memory, and restart data from Prometheus, visualised through interactive time-series charts to detect performance anomalies instantly.



AI Generation: Uses an LLM to generate exploit scenarios against OWASP Juice Shop, cross-referenced with Kube-Hunter scan results to surface cluster misconfigurations.